SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1106

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 60%

QUESTIONS: 6 **Total Marks:60**

Annexure A

Application Based Questions

Code of Conduct:

I D Raghul 192511115 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Application Based Questions

1. Use Case Diagram Analysis

A **Smart Pharmacy Management System** allows customers to search medicines, upload prescriptions, place medicine orders, and make online payments. Pharmacists verify prescriptions and dispense medicines, while administrators manage inventory and user accounts. The current system design does not clearly define actor responsibilities and interactions.

Analyze the system requirements, identify all actors and missing use cases, and design an optimized **Use Case Diagram** showing appropriate relationships (**include, extend, and generalization**) to improve system functionality.

(10 Marks)

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2. Class Diagram Design

A **Smart Library Management System** consists of entities such as Book, Member, Librarian, Reservation, Fine, and BorrowRecord. The existing system does not clearly specify class attributes, operations, or relationships among these entities.

Analyze the system, identify relevant classes with appropriate attributes and methods, define relationships (**association, aggregation, composition, and inheritance**), specify multiplicity, and construct a well-structured **Class Diagram**.

(10 Marks)

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3. Sequence Diagram Evaluation

An **Online Examination System** enables students to log in, select an examination, answer questions, submit responses, and receive results. The communication among the student, examination module, evaluation module, and database is incomplete.

Analyze the interaction process, identify all participating objects and missing message exchanges, and design a detailed **Sequence Diagram** representing the complete interaction flow with activation bars and lifelines.

(10 Marks)

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4. Activity Diagram Construction

A **Loan Approval System** allows customers to submit loan applications, upload supporting documents, complete eligibility verification, receive loan approval or rejection, and complete loan disbursement. The workflow lacks proper decision points and parallel activities.

Analyze the business process, identify all workflow activities, include decision nodes for approval and rejection, represent parallel processing where applicable, and construct an optimized **Activity Diagram**.

(10 Marks)

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5. Component Diagram Design

A **Smart E-Commerce Platform** consists of modules such as User Management, Product Catalog, Shopping Cart, Order Management, Payment Gateway, Inventory Management, and Notification Service. The existing architecture does not clearly define interfaces and dependencies among modules.

Analyze the software architecture, identify major components and their provided/required interfaces, define dependencies among components, and design a comprehensive **Component Diagram** representing the overall system architecture.

(10 Marks)

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6. Deployment Diagram Design

A **Smart Agriculture Monitoring System** uses IoT soil sensors, weather stations, irrigation controllers, cloud servers, mobile applications, and farmer dashboards to monitor crop health and automate irrigation. The deployment architecture lacks proper representation of hardware nodes and communication links.

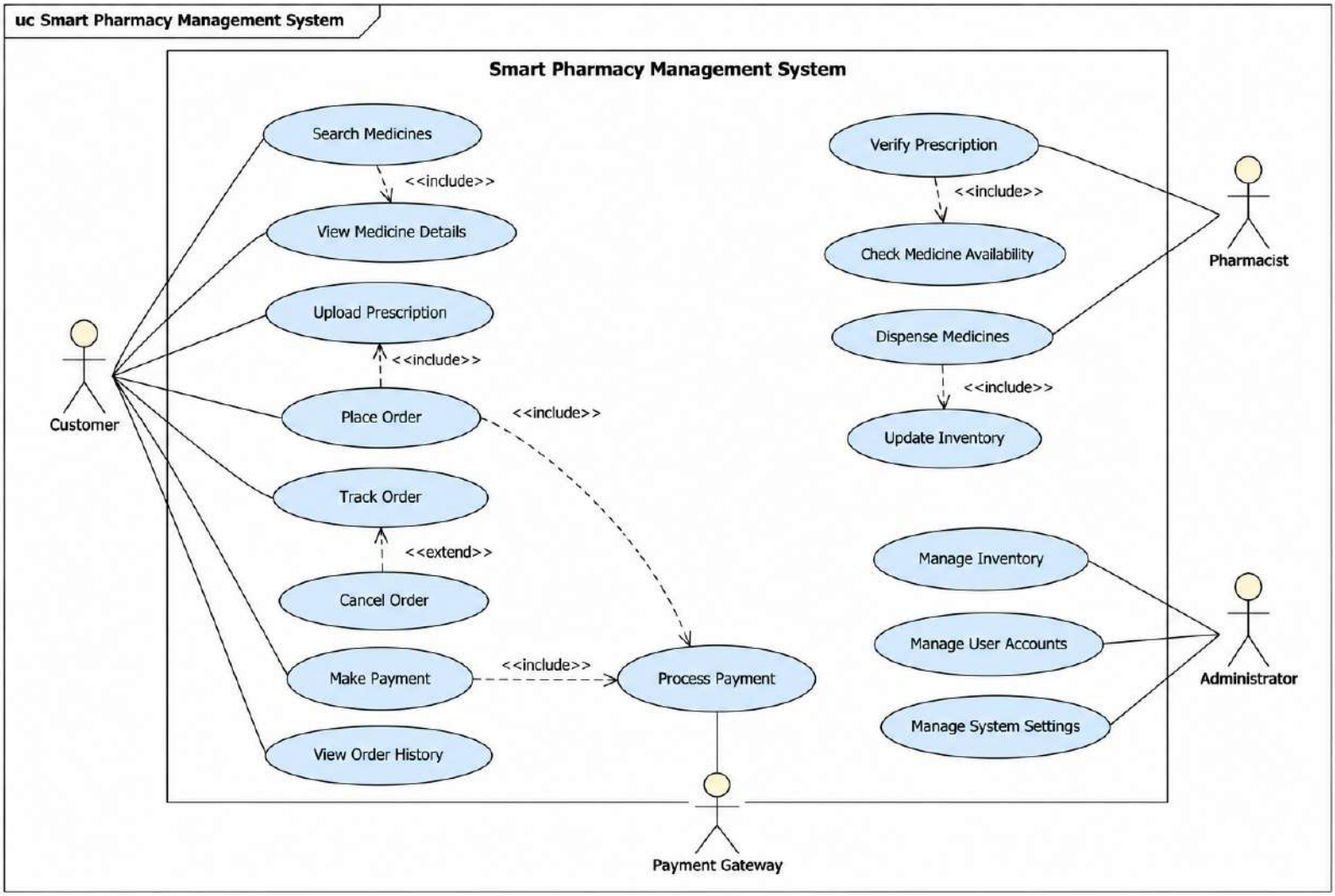
Analyze the deployment environment, identify hardware nodes and deployed software artifacts, specify communication protocols between nodes, and construct a detailed **Deployment Diagram** representing the complete physical system architecture.

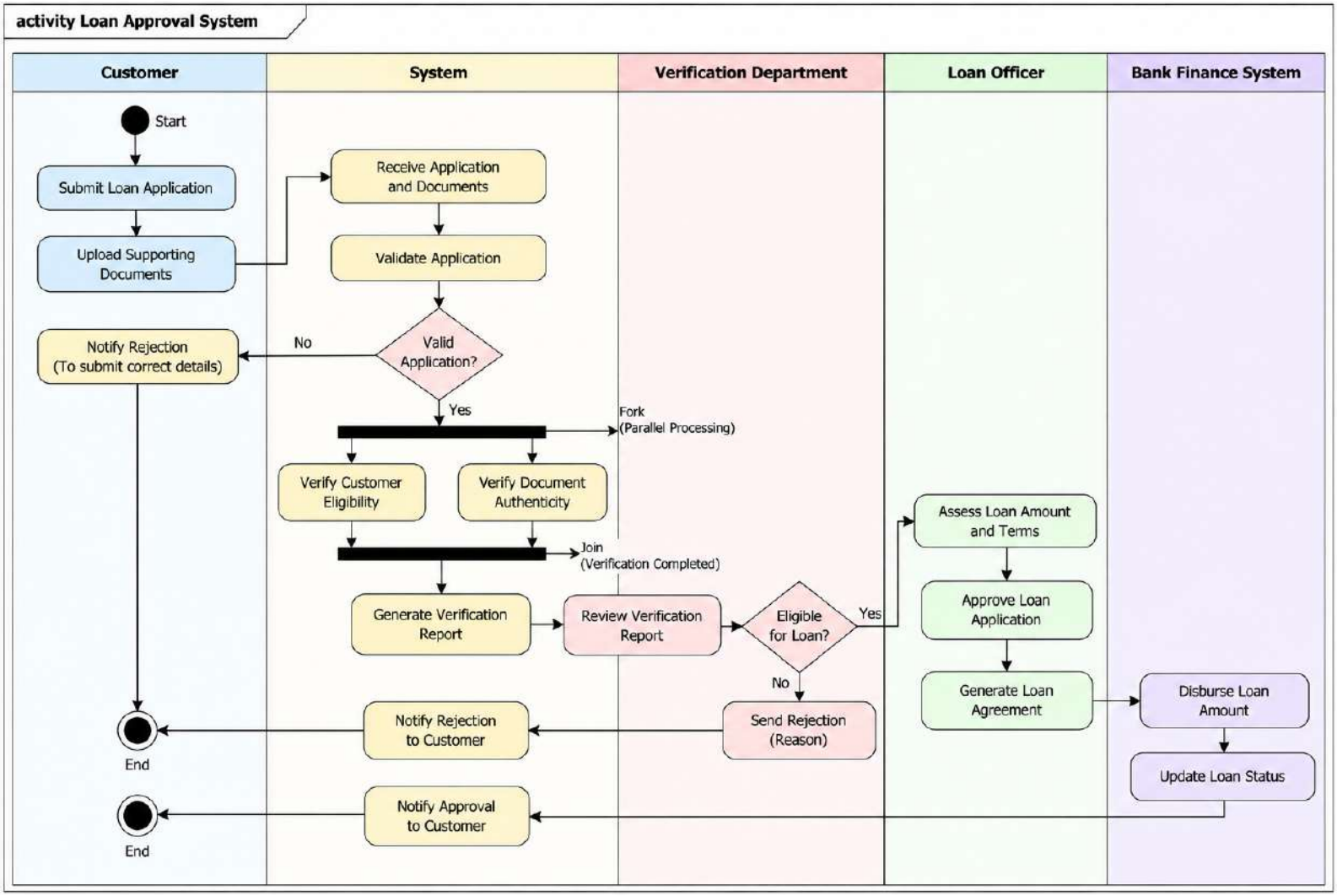
(10 Marks)

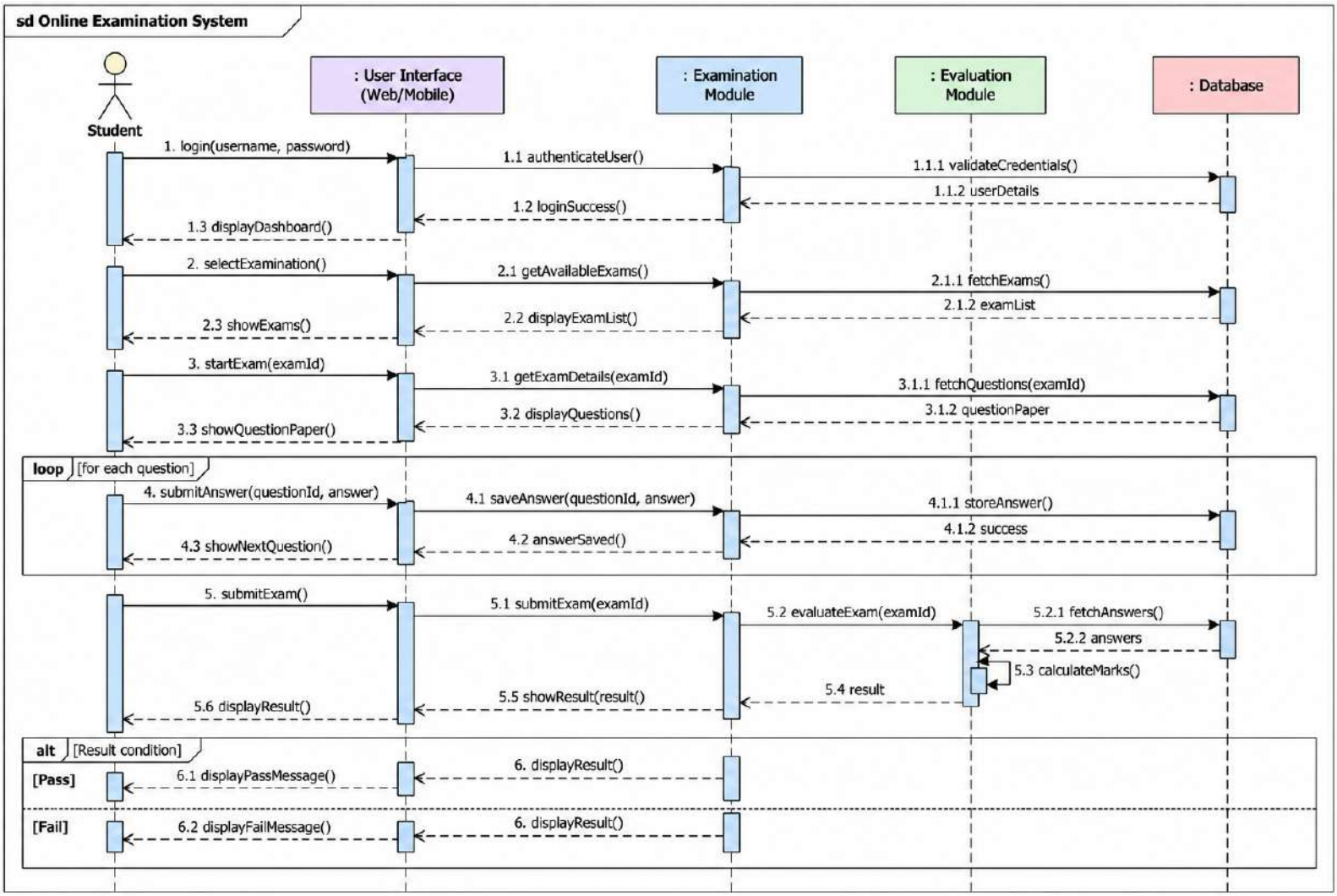
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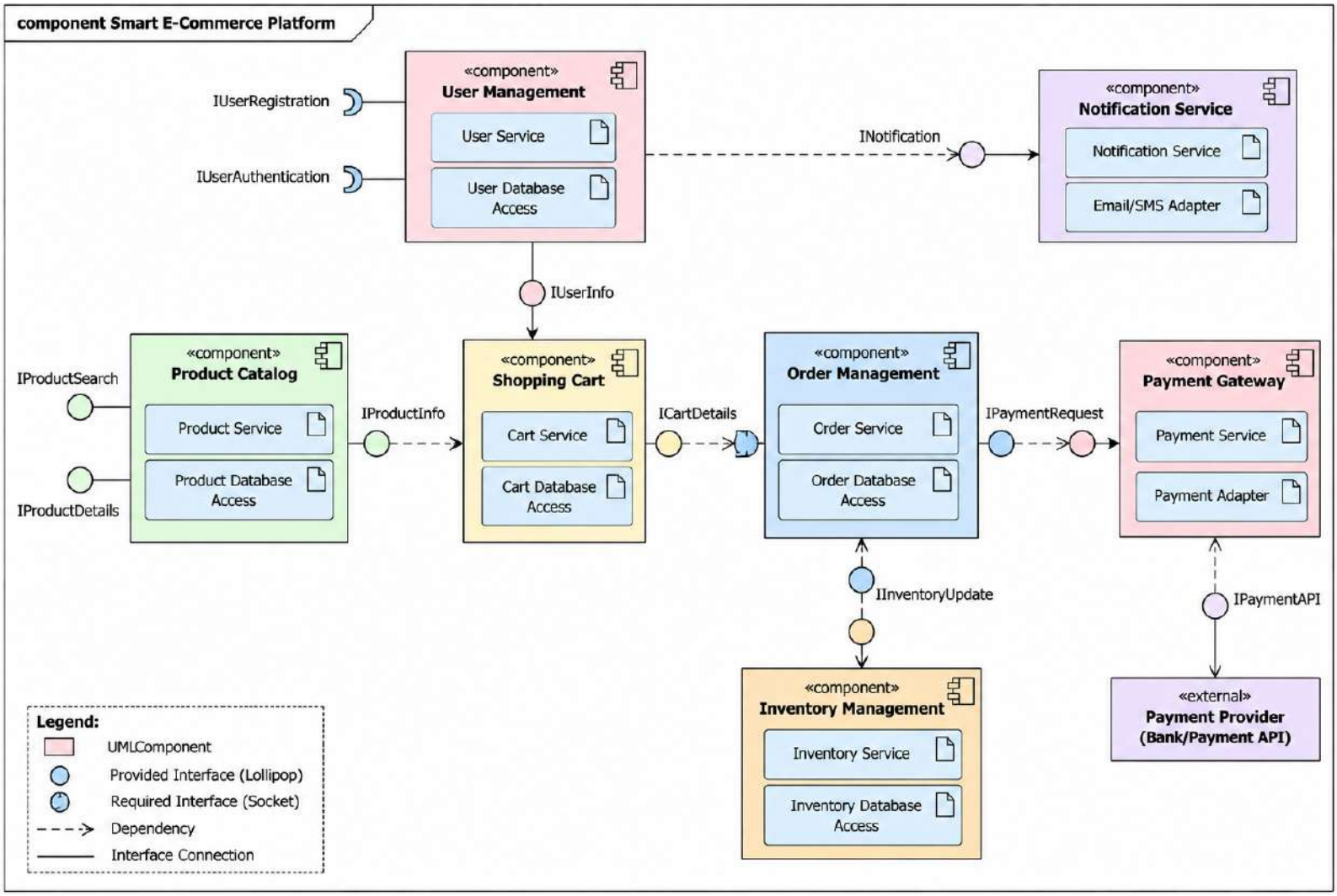
Application Based Questions

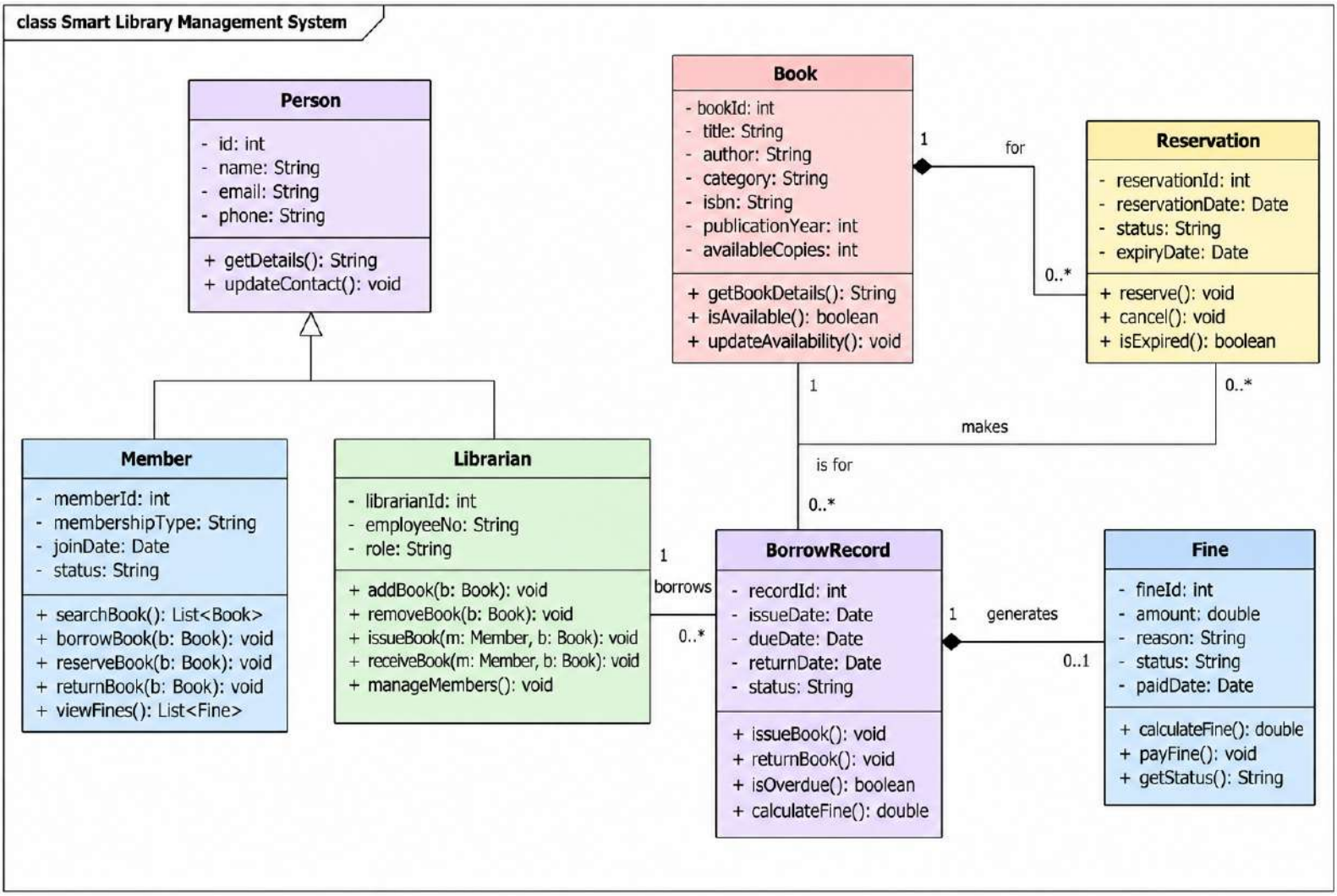
Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

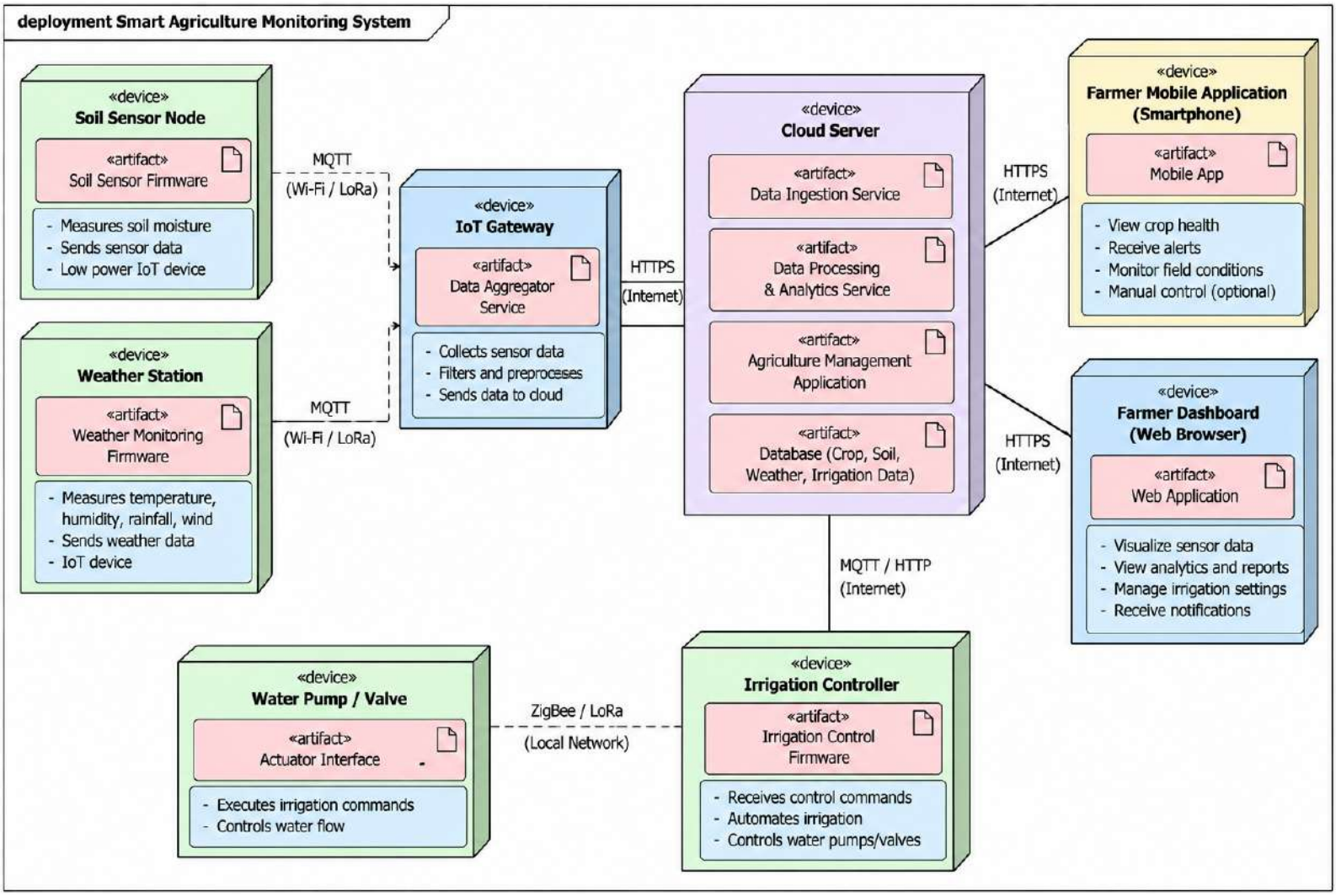

uc Smart Pharmacy Management System
Smart Pharmacy Management System
Search Medicines
View Medicine Details
Upload Prescription
Place Order
Track Order
Cancel Order
Make Payment
View Order History
Process Payment
Verify Prescription
Check Medicine Availability
Dispense Medicines
Update Inventory
Manage Inventory
Manage User Accounts
Manage System Settings
<<include>>
<<include>>
<<include>>
<<include>>
<<extend>>
<<include>>
<<include>>
<<include>>
Customer
Pharmacist
Administrator
Payment Gateway


activity Loan Approval System
Customer
System
Verification Department
Loan Officer
Bank Finance System
Start
Submit Loan Application
Upload Supporting Documents
Receive Application and Documents
Validate Application
Valid Application?
No
Notify Rejection (To submit correct details)
Yes
Fork (Parallel Processing)
Verify Customer Eligibility
Verify Document Authenticity
Join (Verification Completed)
Generate Verification Report
Review Verification Report
Eligible for Loan?
Yes
No
Send Rejection (Reason)
Notify Rejection to Customer
End
Assess Loan Amount and Terms
Approve Loan Application
Generate Loan Agreement
Disburse Loan Amount
Update Loan Status
Notify Approval to Customer
End


sd Online Examination System
Student
: User Interface (Web/Mobile)
: Examination Module
: Evaluation Module
: Database
1. login(username, password)
1.1 authenticateUser()
1.1.1 validateCredentials()
1.1.2 userDetails
1.2 loginSuccess()
1.3 displayDashboard()
2. selectExamination()
2.1 getAvailableExams()
2.1.1 fetchExams()
2.1.2 examList
2.2 displayExamList()
2.3 showExams()
3. startExam(examId)
3.1 getExamDetails(examId)
3.1.1 fetchQuestions(examId)
3.1.2 questionPaper
3.2 displayQuestions()
3.3 showQuestionPaper()
loop [for each question]
4. submitAnswer(questionId, answer)
4.1 saveAnswer(questionId, answer)
4.1.1 storeAnswer()
4.1.2 success
4.2 answerSaved()
4.3 showNextQuestion()
5. submitExam()
5.1 submitExam(examId)
5.2 evaluateExam(examId)
5.2.1 fetchAnswers()
5.2.2 answers
5.3 calculateMarks()
5.4 result
5.5 showResult(result()
5.6 displayResult()
alt [Result condition]
[Pass]
6.1 displayPassMessage()
6. displayResult()
[Fail]
6.2 displayFailMessage()
6. displayResult()


component Smart E-Commerce Platform
«component»
User Management
User Service
User Database Access
IUserRegistration
IUserAuthentication
INotification
«component»
Notification Service
Notification Service
Email/SMS Adapter
IUserInfo
«component»
Product Catalog
Product Service
Product Database Access
IProductSearch
IProductDetails
IProductInfo
«component»
Shopping Cart
Cart Service
Cart Database Access
ICartDetails
«component»
Order Management
Order Service
Order Database Access
IPaymentRequest
«component»
Payment Gateway
Payment Service
Payment Adapter
IInventoryUpdate
IPaymentAPI
«component»
Inventory Management
Inventory Service
Inventory Database Access
«external»
Payment Provider
(Bank/Payment API)
Legend:
UMLComponent
Provided Interface (Lollipop)
Required Interface (Socket)
Dependency
Interface Connection


class Smart Library Management System
Person
- id: int
- name: String
- email: String
- phone: String
+ getDetails(): String
+ updateContact(): void
Book
- bookId: int
- title: String
- author: String
- category: String
- isbn: String
- publicationYear: int
- availableCopies: int
+ getBookDetails(): String
+ isAvailable(): boolean
+ updateAvailability(): void
1
for
0..*
Reservation
- reservationId: int
- reservationDate: Date
- status: String
- expiryDate: Date
+ reserve(): void
+ cancel(): void
+ isExpired(): boolean
1
makes
0..*
is for
0..*
Member
- memberId: int
- membershipType: String
- joinDate: Date
- status: String
+ searchBook(): List<Book>
+ borrowBook(b: Book): void
+ reserveBook(b: Book): void
+ returnBook(b: Book): void
+ viewFines(): List<Fine>
Librarian
- librarianId: int
- employeeNo: String
- role: String
+ addBook(b: Book): void
+ removeBook(b: Book): void
+ issueBook(m: Member, b: Book): void
+ receiveBook(m: Member, b: Book): void
+ manageMembers(): void
1
borrows
0..*
BorrowRecord
- recordId: int
- issueDate: Date
- dueDate: Date
- returnDate: Date
- status: String
+ issueBook(): void
+ returnBook(): void
+ isOverdue(): boolean
+ calculateFine(): double
1
generates
0..1
Fine
- fineId: int
- amount: double
- reason: String
- status: String
- paidDate: Date
+ calculateFine(): double
+ payFine(): void
+ getStatus(): String


deployment Smart Agriculture Monitoring System
«device»
Soil Sensor Node
«artifact»
Soil Sensor Firmware
- Measures soil moisture
- Sends sensor data
- Low power IoT device
MQTT
(Wi-Fi / LoRa)
«device»
Weather Station
«artifact»
Weather Monitoring Firmware
- Measures temperature, humidity, rainfall, wind
- Sends weather data
- IoT device
MQTT
(Wi-Fi / LoRa)
«device»
IoT Gateway
«artifact»
Data Aggregator Service
- Collects sensor data
- Filters and preprocesses
- Sends data to cloud
HTTPS
(Internet)
«device»
Cloud Server
«artifact»
Data Ingestion Service
«artifact»
Data Processing & Analytics Service
«artifact»
Agriculture Management Application
«artifact»
Database (Crop, Soil, Weather, Irrigation Data)
HTTPS
(Internet)
«device»
Farmer Mobile Application (Smartphone)
«artifact»
Mobile App
- View crop health
- Receive alerts
- Monitor field conditions
- Manual control (optional)
HTTPS
(Internet)
«device»
Farmer Dashboard (Web Browser)
«artifact»
Web Application
- Visualize sensor data
- View analytics and reports
- Manage irrigation settings
- Receive notifications
MQTT / HTTP
(Internet)
«device»
Irrigation Controller
«artifact»
Irrigation Control Firmware
- Receives control commands
- Automates irrigation
- Controls water pumps/valves
ZigBee / LoRa
(Local Network)
«device»
Water Pump / Valve
«artifact»
Actuator Interface
- Executes irrigation commands
- Controls water flow